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service to a majority of customers without any failure or disruption of service.

Multimedia

Multimedia, non-linear video editing, and audio editing have had a long-standing relationship with the storage subsystem. Hard disk drives are a critical component to large multimedia design houses, editing workstations and video servers. Advances in drive technology have significantly boosted capacity, performance and reliability and at the same time these larger and speedier drives have become truly affordable. There is no doubt that digital video production and multimedia content creation are amongst the most challenging applications for hard disk drives. The sheer size of digital video files creates unparalleled storage requirements. A half-hour television show digitised for delivery and for broadcast can require up to 45 GB of storage! Through the years, with sustained and continuous incremental improvements, drives have become more and more an integral part of these systems.

Solving just the storage requirements of these applications isn't enough. This is primarily because digital video is a real time format, and the inability to store and retrieve video in real time causes loss of quality due to dropped frames or audio breaks and this is simply unacceptable to digital video producers. There is no way any multimedia content generating producer will tolerate glitches, spikes or audio breaks and the only way to solve

this to a degree is by increasing throughput. Part of this can be solved by increasing the areal density but a great deal depends on the speed at which the drives rotate. Increasing the rpm to 7,200 rpm, 10,000 and as high as 15,000 rpm brings forth some physical limitations and hence higher buffers with ECC are required.

Without the advancements made in



Non-linear video editing apps like Adobe Premiere gobble up disk space faster than you can say 'ouch'!

the field of storage, multimedia on the desktop would not have evolved this rapidly. As the computer moves from a productivity tool to become an entertainment appliance, even the commonest of home users demands a multimedia capable computer. This means that all PCs must now ship with obscenely large hard drives to meet the needs of users, be it for digital music or video.

Conclusion

Of course, all these advancements in speed and storage capacities are inconsequential if the drives don't operate reliably in actual real world operation. With such large files and expensive data, a disk crash can be financially disastrous. Drive manufacturers therefore aggressively pursue reliability during drive design and manufacturing. Each new hard disk drive is designed to achieve specific reliability goals before shipping—each drive must pass performance

Staying Ahead of the Curve

While Seagate is creating efficient new ways of building drives and reducing costs at the component and product level, the underlying technologies that go into disk drives are expected to evolve significantly and swiftly.

Since the drive industry began some 40 years ago, the critical driving metric has been areal density—the number of bits recorded on a square inch of the disk's surface. Disk drive companies are in a relentless race to increase areal density. Areal density has increased since the world's first disk drive in the late 1950s by over 10 million times—an improvement rate unparalleled by any industry in history! Today, areal density is increasing at about 100 per cent per year; nearly two times the rate of increase seen only a couple of years ago. To ensure that Seagate holds the lead position in this critical race, it has dramatically increased technical spending for head and disk R&D. Since 1996, Seagate has increased spending for recording head R&D by three times and for disks by nearly five times. Focused efforts by the Seagate team have the company jockeying for first place, as recent areal density demonstrations have shown.

and stress tests at the extreme parameters of the drive's specifications, including tests for shock, temperature and operating environment. At the same time they are constantly working on ways to reduce the number of moving components in the drive, thereby reducing the risk of failure substantially. This dual focus on producing more reliable drives and monitoring critical operating parameters have combined to allow hard drives to meet the stringent uptime requirements of today's applications.

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